



Answer the following questions

Question 1:

[16 Marks]

A) Answer the following questions about Support Vector Machines (SVMs)

[8 Marks]

- i. SVM are best suited to solve linear classification problems, explain how SVM can be adapted to deal with the case of non-linear data.
- ii. State whether the following statements are either **True** or **False**.
 1. The support vectors which are the data points that lie closest to the decision surface are the easiest ones to classify.
 2. When you remove one of the support vector points, the decision boundary will change.
 3. Two SVM classifier models will be needed to train when dealing with four-class classification problem using one-vs-all method.
 4. The kernel function of the SVM maps low dimensional data to high dimensional space.
 5. The SVM is considered an efficient classifier for large sized data sets.

B) Background subtraction takes the (absolute) difference between a background image frame $B(x, y, t)$ and an image frame $I(x, y, t)$, and determines foreground pixels by applying a threshold on the result.

[8 Marks]

- i. Explain two methods where the background image $B(x, y, t)$ can be estimated.
- ii. What are the disadvantage(s) of each method (if exist)?

Question 2:

[16 Marks]

A) **RANdom SAMple Consensus (RANSAC)** is an algorithm for fitting a model to very noisy data. Its basic idea is to fit models to small samples of data, and then look for "consensus" from the entire dataset.

Define the used parameters of the **RANSAC** algorithm, and how the value of each parameter should be selected.

[8 Marks]

B) In foreground-background image segmentation, the **Grabcut** is considered one of the fundamental approaches. Answer the following questions about **Grabcut**

[8 Marks]

- i. What image features can be used for segmentation?
- ii. How initial segmentation of image pixels can be done?
- iii. What kind of model representation can be formulated to define both foreground and background classes?
- iv. What is the main technique upon which the Grabcut approach depends for graph partitioning?

Question 3:**[18 Marks]**

A) Describe the **Bag-Of-visual-Words (BOW)** image representation technique, stating what is meant by a visual word. Show how the **BOW** can be used for the problem of object recognition. [7 Marks]

B) The **Lucas-Kanade (LK)** method is used for estimating the displacement (optical flow) of an image patch between image frames. [11 Marks]

- i. State what are the brightness constancy constraint equation and its assumptions?
- ii. Define the aperture problem that occurs in the field of visual motion. In what way does the brightness constancy constraint equation exhibit the aperture problem?
- iii. How the **LK** method is used to solve the aperture problem? What assumption(s) it is based on.
- iv. The **LK** method assumes small displacements between frames. Explain how large displacements are handled in the **LK** method.

Question 4**[15 Marks]**

A) A feature descriptor is a representation of an image that simplifies the image by extracting useful information and throwing away extraneous information. Typically, a feature descriptor converts an image of size width x height x 3 (channels) to a feature vector / array of length n. [11 Marks]

- i. State what are the characteristics of a good feature descriptor?
- ii. Explain the steps of the **Histogram of Oriented Gradients (HOG)** feature descriptor.

B) Assume that you have a video of the street filmed from a static camera where there is a car you need to track. Assume that you have a color model of the car that you are tracking. The model is described as a color histogram that has 16 equally-spaced bins of (only) the hue channel in HSV color space. Explain how this color model can be used to track the car along the sequence of frames in the video. [4 Marks]

With Best Regards,

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